Non-Ergodic Mess3 Mixtures: How Transformers Represent Hierarchical Belief Structure

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Abstract

We train a small transformer on a non-ergodic dataset of Mess3 processes with different parameters, where each sequence comes from one ergodic component. The model must solve a hierarchical inference problem: identifying the component (meta-synchronization) and tracking belief states within it. We derive geometric predictions connecting this to the Factored World Hypothesis, verify Sierpinski gasket recovery ($R^2 = 0.989$), and analyze synchronization dynamics across positions and layers.

1 Introduction

Language models face non-ergodic data: different documents constitute distinct statistical regimes. In-context learning requires identifying the active regime before predicting optimally. We study this in a controlled setting using Mess3 processes (**??**), 3-state nonunifilar HMMs whose belief geometry forms a Sierpinski gasket.

By constructing a non-ergodic training set where each sequence comes from one of K Mess3 parameterizations, we create a minimal in-context learning problem: the transformer must determine which process generated the sequence and track its hidden state. This connects to the Factored World Hypothesis (FWH) (**?**): if the model treats component identity and within-component belief as separate factors, it should represent them in orthogonal subspaces with dimensionality $(K-1)+2 = K+1$, rather than the joint $3K-1$. Crucially, our non-ergodic mixture has *block-diagonal* transition operators (not tensor-product), so the standard FWH does not directly apply. This creates an interesting intermediate case: does the factoring inductive bias extend beyond conditionally-independent processes? The FWH predicts that transformers first learn factored representations, expanding dimensionality only under sustained gradient pressure (**?**). Our setup tests whether this pattern holds for block-diagonal structure.

Contributions. (1) We formalize the non-ergodic Mess3 mixture as hierarchical belief inference and derive geometric predictions, identifying three possible geometries (§4). (2) We verify single-process gasket recovery ($R^2 = 0.989$) and compare model loss against the Bayes-optimal predictor at each position (§5). (3) We analyze synchronization dynamics: how quickly the model identifies the active component across context positions and layers (§6).

2 Background

Mess3 process. A 3-state nonunifilar HMM emitting $z \in \{0, 1, 2\}$, parameterized by emission clarity α and state-switching rate x, with $\beta = (1-\alpha)/2$, $y = 1-2x$. The labeled transition matrices $T^{(z)}_{ij} = P(\text{emit } z, \text{go to } j \mid \text{state } i)$ are given in Appendix A. The process is nonunifilar: observing a token does not uniquely determine the state transition, so optimal prediction requires maintaining a belief distribution over hidden states. These belief states form a Sierpinski gasket in the 2-simplex (**?**). The subdominant eigenvalue $\zeta = 1-3x$ controls the exponential decay rate of belief corrections with context distance. The entropy rate $H = -\sum_s \pi(s) \sum_z P(z|s) \log P(z|s)$ with uniform $\pi(s) = 1/3$ gives the theoretical minimum cross-entropy.

Belief geometry in transformers. Transformers linearly represent belief states $\boldsymbol{\eta}^{(z_{1:t})} = \boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)}/\text{norm}$ in their residual stream (**?**). This representation is a constrained Bayesian approximation: $r_1(z_{1:d}) = \boldsymbol{\pi} + \sum_{s=1}^{d} (\boldsymbol{\pi} T^{z_s} T^{d-s} - \boldsymbol{\pi})$, where each correction decays as ζ^{d-s}, reflecting the parallel nature of attention (**?**).

Factored World Hypothesis. Beyond single processes, transformers have an inductive bias toward factored representations (**?**): conditionally

independent factors are represented in orthogonal subspaces, requiring $\sum_n (d_n - 1)$ dimensions vs. $\prod_n d_n - 1$ (joint). In synthetic experiments with 5 factors (3 Mess3 + 2 Bloch Walk, joint vocab 432), activations converge to 10 dimensions (factored) rather than 242 (joint). This factoring persists even when it sacrifices fidelity, acting as a representational attractor that models depart from only under sustained gradient pressure.

ICL as source discrimination. Non-ergodicity forces hierarchical inference: identify the source, then predict within it. This is structurally identical to in-context learning in LLMs (**?**). ICL and activation steering both modify the model's belief state about latent concepts, operating additively in log-belief space (**?**).

3 Non-Ergodic Mess3 Mixture

3.1 Construction

We construct K Mess3 processes with distinct parameters $\{(\alpha_k, x_k)\}_{k=1}^{K}$. Each training sequence is generated entirely by one component chosen uniformly; the model observes only tokens, not the component identity.

We choose components with well-separated entropy rates H_k and subdominant eigenvalues $\zeta_k = 1 - 3x_k$:

	α	x	H_k (nats)	ζ_k
C0	0.95	0.03	0.41	0.91
C1	0.85	0.05	0.69	0.85
C2	0.65	0.15	1.03	0.55

Table 1: Mess3 components. H_k: entropy rate (minimum CE). ζ_k: subdominant eigenvalue (controls belief convergence rate). C0 is highly predictable with slow mixing; C2 is nearly uniform with fast mixing.

3.2 Relevance to Language Models

This structure mirrors natural language in three ways. **ICL as source discrimination**: the model must use early tokens to identify the generating process, analogous to topic/style adaptation. **Covariate shift**: different components have different token distributions, requiring in-context adaptation. **Hierarchical latent structure**: component identity is a slow variable (constant per sequence), hidden state is fast (evolves per token), mirroring the multi-scale structure of language.

3.3 Experimental Setup

Model. GPT-2 style decoder-only transformer via TransformerLens (**?**). Phase 1 (single Mess3): 3 layers, $d_{\text{model}} = 64$, 4 heads. Phase 2 (mixture): 4 layers, $d_{\text{model}} = 128$, 4 heads. Pre-norm, learned positional embeddings, vocabulary $\{$BOS, 0, 1, 2$\}$.

Training. Next-token CE loss, Adam (lr $= 5 \times 10^{-4}$, no weight decay). Fresh batches of 8192 sequences generated on-the-fly. Context: 16 tokens + BOS. Phase 1: 10K steps, Phase 2: 5K steps.

Baselines. Bayes-optimal CE at each position via Monte Carlo: both *oracle* (knows component) and *Bayesian* (infers component from observations).

Analysis. (1) CEV via PCA, (2) linear regression to belief states (R^2), (3) meta-belief regression, (4) per-position/layer probing for synchronization dynamics, (5) attention patterns.

4 Pre-Registered Predictions

4.1 Mathematical Derivation

The joint state space $(k, s) \in \{1, \ldots, K\} \times \{1, 2, 3\}$ has $3K$ states. Since component identity k is fixed within each sequence, the predictive vector factorizes:

$$P(k, s \mid z_{1:t}) = P(k \mid z_{1:t}) \cdot P(s \mid k, z_{1:t}) \quad (1)$$

The meta-belief $P(k \mid z_{1:t})$ is the posterior over components; $P(s \mid k, z_{1:t})$ is the within-component belief. The joint transition operator is block-diagonal: $T^{(z)} = \bigoplus_k T_k^{(z)}$, with $P(k)$ entering only through the prior/posterior update. However, k is static: after sufficient context $P(k \mid z_{1:t})$ concentrates on the true component.

The meta-belief evolves as $P(k \mid z_{1:t}) \propto P(k) \prod_{\tau=1}^{t} P(z_\tau \mid z_{1:\tau-1}, k)$, where concentration rate depends on KL divergence between component predictive distributions.

4.2 Geometric Predictions

Prediction 1: Dimensionality. Under a factored representation: $(K-1)$ dims for the meta-belief simplex + 2 dims for within-component belief = $K+1$ total. For K=3: factored = 4 dims, joint = 8 dims, separated gaskets $\approx$ 6 dims.

Prediction 2: Context-position dependence. At t=1, meta-belief is near-uniform: geometry is a

superposition of K gaskets. As t increases, meta-belief sharpens, geometry resolves into component-specific clusters. Per-position loss should decrease toward the Bayes-optimal rate.

Prediction 3: Layer dependence. Embedding: 3 token vertices. Early layers: component discrimination emerges (meta-synchronization). Later layers: within-component gasket structure sharpens. This ordering reflects computational dependency: identify the process before applying correct transition matrices.

4.3 Multiple Possible Geometries

1. **Separated gaskets**: K distinct gaskets with large offsets encoding component identity ($\sim 2K$ dims). Natural for block-diagonal structure.

2. **Factored (shared belief subspace)**: component identity in $(K-1)$D, shared 2D belief subspace with superimposed gaskets ($K+1$ dims). Strongest evidence for FWH extending beyond tensor-product structure.

3. **Conditional (component-specific belief subspaces)**: each component gets its own oriented 2D subspace ($2K+K-1$ dims).

We predict geometry 1 or 2. Following the inductive bias observed in (**?**), the model may initially learn geometry 1, transitioning toward 2 over training as the factoring attractor pulls representations toward dimensional efficiency.

5 Results

5.1 Phase 1: Single Mess3 Verification

We first verify our setup on a single Mess3 ($\alpha = 0.85$, $x = 0.05$; 3 layers, $d_{\text{model}} = 64$, 10K steps). Training loss converges to 0.82 nats, matching the Bayes-optimal CE of 0.82. Linear regression (held-out 80/20 split) from final-layer activations to belief states achieves $R^2 = 0.989$ (all positions aggregated; $R^2 = 0.947$ at the last position only). The Sierpinski gasket is clearly recovered (Figure 1, left column is analogous). Effective dimensionality decreases from 17 to 6 over training.

5.2 Phase 2: Non-Ergodic Mixture

We train a 4-layer model ($d_{\text{model}} = 128$) on the 3-component mixture (Table 1) for 5K steps (loss converges by step $\sim$2K).

Belief regression. All three component-specific gaskets are recovered with high fidelity (Table 2). The meta-belief (posterior over components) is linearly decodable with $R^2 = 0.941$. Figure 1 shows ground-truth vs. predicted gaskets.

	R^2 (held-out)	RMSE
C0 (α=0.95, x=0.03)	0.996	0.029
C1 (α=0.85, x=0.05)	0.991	0.039
C2 (α=0.65, x=0.15)	0.989	0.026
Meta-belief	0.941	0.046

Table 2: Belief regression (held-out 20% test set). Within-component beliefs are near-perfectly decoded; meta-belief is also well-represented.

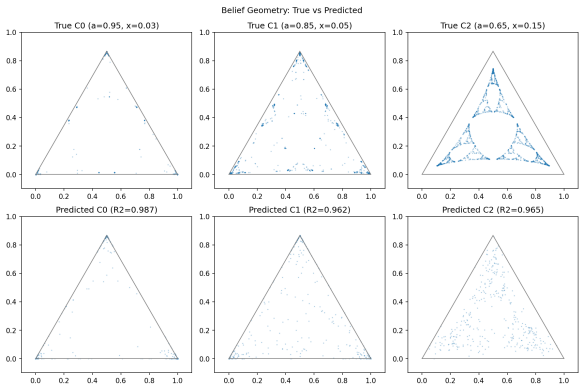

Figure 1: Ground-truth gaskets (top) vs. predicted from residual stream (bottom). Each component produces a distinct Sierpinski gasket, all recovered by linear regression.

Dimensionality. Effective dimensionality (95% CEV) converges to 6, between the factored prediction ($K+1 = 4$) and joint prediction ($3K-1 = 8$). This suggests partial factoring with additional dimensions for positional encoding.

Per-position loss. The model tracks the Bayesian baseline closely (Figure 2), remaining above the oracle baseline. The gap quantifies the cost of imperfect meta-synchronization. Near-optimal inference is achieved by position $\sim$6.

6 Additional Analysis: Synchronization Dynamics

We chose synchronization dynamics because it directly probes the hierarchical structure of in-context learning: does the model first identify the source, then refine beliefs?

Layer-wise probing. Figure 3 shows held-out R^2 for meta-belief and within-component belief probes

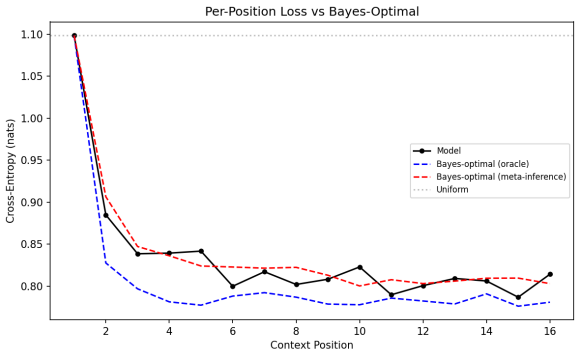

Figure 2: Per-position CE: model (black) tracks Bayesian baseline (red, infers component) closely, above oracle (blue, knows component).

at each layer. Within-component belief is already $R^2 = 0.86$ at embedding and reaches 0.99 at L0. Meta-belief (component identity) starts near zero and builds progressively: $0.00 \rightarrow 0.48 \rightarrow 0.86 \rightarrow 0.93 \rightarrow 0.94$ across embed→L0→L1→L2→L3.

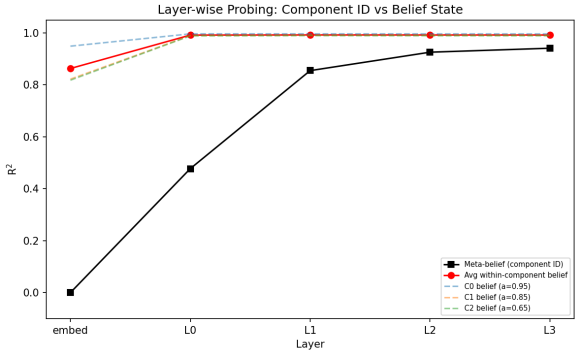

Figure 3: Within-component belief (red) is nearly perfect from L0; meta-belief (black) builds progressively across layers.

This is surprising: we predicted component identification before belief refinement (Prediction 3), but the model maintains parallel belief tracks for all components, with the correct track becoming dominant as meta-synchronization sharpens across layers. This is consistent with the constrained belief update: attention corrections are valid regardless of which component is active.

Position-wise probing. The position-wise meta-belief R^2 shows a different pattern: it is near-perfect at positions 2–4 ($R^2 \approx 1.0$) and gradually *decreases* to 0.94 by position 16. This occurs because early positions have only 3 possible belief states (one per token value), making linear separation trivial. As context grows, belief geometry becomes richer and the linear probe faces a harder task. The meaningful dimension of meta-synchronization is therefore *across layers* (where it genuinely emerges), not across positions.

What we would do with more time. (1) Causal interventions: perturb the component-identity direction and verify factored independence. (2) Scaling with K: test whether dimensionality follows $K+1$. (3) GHMM extension: add Bloch Walk components via belief-state-based generation. (4) Training dynamics: track when separated vs. factored geometries emerge, testing whether the FWH attractor applies to block-diagonal structure.

7 Discussion

Non-ergodicity forces hierarchical inference. The model cannot achieve low loss without first identifying the source. This mirrors how LLMs must identify the active regime before predicting well within it.

Block-diagonal structure. The joint operator is block-diagonal, not tensor-product, placing our setup outside the standard FWH setting. Whether the factoring inductive bias extends to this case is an open question that our experiments address.

Residual stream as belief space. The residual stream has no privileged basis; geometric arrangement encodes prediction similarity. If the model discovers that source identity and within-source state are independent dimensions of prediction, it should allocate orthogonal subspaces accordingly.

Limitations. Small model (3–4 layers); static components (no drift or switching); same token alphabet across components. Natural extensions include slowly drifting parameters, GHMM components such as Bloch Walk (which would require belief-state-based generation rather than hidden state simulation), and connections to induction heads (**?**) as a mechanism for source discrimination.

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A Mess3 Transition Matrices

With $\beta = (1-\alpha)/2$ and $y = 1 - 2x$:

$$T^{(0)} = \begin{pmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{pmatrix} \tag{2}$$

$$T^{(1)} = \begin{pmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{pmatrix} \tag{3}$$

$$T^{(2)} = \begin{pmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{pmatrix} \tag{4}$$

The net transition matrix is:

$$T = T^{(0)} + T^{(1)} + T^{(2)} = \begin{pmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{pmatrix} \tag{5}$$

with uniform stationary distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$ and non-trivial eigenvalue $\zeta = 1 - 3x$.

The emission probability $P(z \mid s) = \sum_j T^{(z)}_{sj}$ gives the entropy rate:

$$H = -\sum_{s=1}^{3} \frac{1}{3} \sum_{z=0}^{2} P(z|s) \log P(z|s) \tag{6}$$

B Bayes-Optimal Loss Derivation

The Bayes-optimal loss at position t is the cross-entropy achieved by a predictor maintaining exact belief states:

$$\mathrm{CE}(t) = \mathbb{E}_{z_{1:t}} \left[-\log P(z_t \mid z_{1:t-1}) \right] \tag{7}$$

where $P(z_t \mid z_{1:t-1}) = \boldsymbol{\eta}^{(z_{1:t-1})} T^{(z_t)} \mathbf{1}$.

For the non-ergodic mixture, we compute two baselines:

- **Oracle**: knows the component, uses component-specific transition matrices.
- **Bayesian**: must infer the component, uses the mixture prediction $P(z_t \mid z_{1:t-1}) = \sum_k P(k \mid z_{1:t-1}) \cdot P(z_t \mid z_{1:t-1}, k)$.

The gap between oracle and Bayesian baselines quantifies the cost of not knowing the component identity.

C Code

All code is available at: https://github.com/seonglae/mats-mess3. The implementation uses PyTorch with TransformerLens for activation extraction. Key files:

- `src/mess3.py`: Mess3 process generation and belief computation
- `src/bayes_optimal.py`: Bayes-optimal loss computation
- `src/sync_analysis.py`: Synchronization dynamics analysis
- `run.py`: Main experiment script